

IPO & IS (Main) – 2016 – Set – 2

Technical Paper

Time : 2 hours

Full Marks : 200

Each question carries 2 marks.

There is negative marking @ 0.5 mark for each wrong answer.

Answer **all** questions, choosing the correct **one** from the alternatives suggested and darken the appropriate circle using **BLUE or BLACK BALL POINT PEN**.

1. The DC resistance of a crystal diode is _____ its AC resistance.
(1) ☒ The same as
(2) ☐ More than
(3) ☒ Less than
(4) ☐ Twice
(5) ☐ Thrice
2. A zener diode is always _____ connected.
(1) ☒ Reverse
(2) ☐ Forward
(3) ☐ Reverse or Forward
(4) ☒ Upward
(5) ☐ Downward
3. The ripple factor of a half wave rectifier is _____.
(1) ☐ 2
(2) ☐ 0.5
(3) ☐ 3
(4) ☐ 1.21
(5) ☐ 2.12
4. A half wave rectifier has an input voltage of 240V rms. If the step down transformer has a turns ratio of 8 : 1, what is the peak load voltage, if diode drop is ignored ?
(1) ☐ 27.5 V
(2) ☒ 86.5 V
(3) ☐ 30.5 V
(4) ☒ 42.5 V
(5) ☐ 62.5 V

5. When negative voltage feedback is applied to an amplifier, its voltage gain _____.

- (1) Is increased
- ☒ (2) Is reduced
- (3) Remains the same
- (4) May increase or reduce
- ☒ (5) Unpredictable

6. The gain of the amplifier without feedback is 100 db. If a negative feedback of 3 db is applied, the gain of the amplifier will become _____.

- ☒ (1) 97 db
- (2) 103 db
- ☒ (3) 300 db
- (4) 53 db
- (5) 153 db

7. An oscillator employs _____ feedback.

- ☒ (1) Positive
- (2) Negative
- ☒ (3) No feedback
- (4) Both (1) and (2)
- (5) Either (1) or (2)

8. The phase difference between the output and input voltages of a CE amplifier is _____ degree.

- (1) 90
- ☒ (2) 270
- (3) 0
- ☒ (4) 180
- (5) 120

9. Which of the following has been assigned as zero octane number?

- (1) Isooctane
- ☒ (2) n-heptane
- ☒ (3) Hexadecane
- (4) TEL
- (5) n-octane

10. CNG stands for:

- (1) Common Natural Gas
- (2) Compressed Normal Gas
- (3) Common Normal Gas
- ☒ (4) Compressed Natural Gas
- (5) None of the above

11. Which of the following metal could provide cathodic protection to iron ?

- (1) ~~Zn~~
- (2) Ni
- (3) Cu
- (4) Fe
- (5) All of the above

12. Which of the following electromagnetic radiation has maximum frequency ?

- (1) X-ray
- (2) ~~Cosmic ray~~
- (3) Microwave
- (4) Ultraviolet
- (5) Spectral

13. Which of the following factors enhance the process of corrosion ?

- (1) Air and moisture
- (2) Presence of strain in the metal
- (3) Presence of Electrolyte

(4) ~~Both (1) and (2)~~

(5) All of the above

14. The electrode potential of SHE (Standard Hydrogen Electrode) is :

- (1) +1 V
- (2) -1 V
- (3) ~~0 V~~
- (4) +1.5 V
- (5) -1.5 V

15. The correct relationship between

E_{cell}^0 and K_{eq} is :

- (1) $E_{\text{cell}}^0 = K_{\text{eq}}$
- (2) $E_{\text{cell}}^0 = -K_{\text{eq}}$
- (3) $K_{\text{eq}} = \frac{0.0591 \log E_{\text{cell}}^0}{n}$
- (4) ~~$E_{\text{cell}}^0 = \frac{0.0591 \log K_{\text{eq}}}{n}$~~
- (5) None of the above

(Turn over)

16. Which of the following is paramagnetic ?

- (1) O_2^+
- (2) N_2
- (3) O_2^{++}
- (4) O_2^{--}
- (5) C_2^{++}

17. Which of the following is TRUE for a spontaneous process ?

- (1) $\Delta G > 0$
- (2) $\Delta G < 0$
- (3) $\Delta G = 0$
- (4) $\Delta H > 0$ and $\Delta S < 0$
- (5) $\Delta H < 0$ and $\Delta S > 0$

18. In an isochoric process _____ remains constant.

- (1) Pressure
- (2) Temperature
- (3) Volume
- (4) Heat change
- (5) Flux

19. Which of the following is NOT an organometallic compound ?

- (1) CH_3COONa
- (2) CH_3Li
- (3) $(\text{CH}_3\text{CH}_2)_2\text{Zn}$
- (4) CH_3MgBr
- (5) CH_2N_3

20. The unit of rate of a chemical reaction is :

- (1) Moles lit^{-1}
- (2) Moles sec^{-1}
- (3) Moles lit sec^{-1}
- (4) Moles $\text{lit}^{-1} \text{sec}^{-1}$
- (5) Moles lit

21. If a body is subjected to plastic impact :

- (1) Only kinetic energy is conserved
- (2) Only momentum is conserved
- (3) Both (1) and (2)
- (4) Strain energy is conserved
- (5) None of the above

22. Frictional force depends on :

- (1) Surface area in contact
- ☒ (2) Roughness of surface
- (3) Both (1) and (2)
- (4) Surface texture
- (5) None of the above

23. A force of 150 N moves a body in 5 sec along a straight line. The equation of motion of the body is $x = t^3 - 60t$, where x is the distance and t is the time. What is the mass of the moving body ?

- (1) 7.23 kg
- (2) 15.23 kg
- (3) 15 kg
- ☒ (4) 5 kg
- (5) 6.5 kg

24. Which of the following laminas do not have centroid at its geometrical centre ?

- (1) Circle

(2) Equilateral triangle

☒ (3) Right angled triangle

(4) Square

(5) None of the above

25. If a material has no uniform density throughout the body, then the position of centroid and center of mass are :

(1) Identical

☒ (2) Not identical

(3) Independent upon the density

(4) Unpredictable

(5) Dependent on mass of the material

26. A block sliding down in an inclined plane has acceleration _____ acceleration due to gravity.

☒ (1) Less than

(2) Greater than

(3) Same as

(4) None of the above

(5) Acceleration may be less or more

27. State the wrong statement about truss :

- (1) Load is applied at the joint of members
- (2) Bending is allowed in the members of the truss
- (3) Both (1) and (2)
- (4) Truss members carry axial loads
- (5) None of the above

28. What is the length of a train which crosses a bridge of 150 m in 20 sec with a speed of 40 km/h ?

- (1) 222 m
- (2) 150 m
- (3) 72.2 m
- (4) 70.5 m
- (5) 75.5 m

29. A ball is dropped on a horizontal surface from a height of 15 m. What will be the rebounding height of the ball after striking the surface ? (Coefficient of restitution = 0.8)

- (1) 9.6 m

- (2) 6.5 m
- (3) 5.6 m
- (4) 6.9 m
- (5) 8.5 m

30. If coefficient of restitution is zero during plastic impact, then _____

- (1) Kinetic energy is conserved
- (2) Momentum energy is conserved
- (3) Bodies move away after impact
- (4) Static energy is conserved
- (5) All types of energy are conserved

31. If two objects of 30 kg and 10 kg move with equal kinetic energy, then what is the ratio of magnitudes for linear momentum ?

- (1) $\sqrt{3} : 1$
- (2) $1 : \sqrt{3}$
- (3) $1 : 3\sqrt{3}$
- (4) $1 : 3$
- (5) $3 : 1$

32. The radius of curvature of trajectory for a profile is minimum, if _____.

- (1) Velocity is minimum
- (2) Acceleration is maximum
- ☒ (3) Both (1) and (2)
- (4) Velocity is maximum
- (5) Acceleration is minimum

33. The force for which work done is independent of _____ is called as conservative force.

- (1) Distance
- ☒ (2) Path
- (3) Time
- (4) Velocity
- (5) Acceleration

34. If a solid sphere, disc and cylinder are allowed to roll down an inclined plane from the same height :

- (1) Cylinder will reach the bottom first
- (2) Disc will reach the bottom first
- ☒ (3) Sphere will reach the bottom first

(4) All will reach the bottom at the same time

(5) Cylinder will reach the bottom last

35. A mixture of gas expands from 0.03m^3 to 0.06m^3 at a constant pressure of 10 bar and absorbs 84 kJ of heat during the process. The change in internal energy of the mixture is :

- (1) 30 kJ
- ☒ (2) 54 kJ
- (3) 84 kJ
- (4) 70 kJ
- (5) 74 kJ

36. The COP of a refrigerator on a reversed Carnot cycle is 5. The ratio of higher absolute temperature to the lower temperature (i. e. T_1/T_2) is :

- (1) 1.25
- (2) 1.3
- (3) 1.4
- ☒ (4) 1.2
- (5) 1.5

37. In an adiabatic process, 5000J of work is performed on a system. The system returns to its original state while 1000J of heat is added. The work done during the non-adiabatic process is :

- (1) + 4000J
- ☒ (2) - 4000J
- (3) + 6000J
- (4) - 6000J
- (5) + 3000J

38. A small steam whistle (perfectly insulated and doing no shaft work) causes a drop of 0.8 kJ/kg in enthalpy of steam from entry to exit. The kinetic energy of the steam at entry is negligible ; the velocity of steam at exit is :

- (1) 4 m/s
- ☒ (2) 40 m/s
- (3) 80 m/s
- (4) 120 m/s
- (5) 140 m/s

39. Steam enters an adiabatic turbine operating at steady state with an enthalpy of 3251.0 kJ/kg respectively. The mass flow rate of steam is 10 kg/s. Kinetic and

potential energy changes are negligible. The power output of the turbine in MW is :

- (1) 6.5
- ☒ (2) 8.9
- (3) 9.1
- (4) 27
- (5) 32

40. A 1 kg block of ice at 0°C is placed into a perfectly insulated, sealed container that has 2 kg of water also at 0°C. The water and ice completely fill the container, but the container is flexible. After some time one can expect that :

- (1) The water will freeze so that the mass of the ice will increase
- ☒ (2) The ice will melt so that the mass of the ice will decrease
- (3) Both the amount of water and the amount of ice will remain constant
- ☒ (4) Both the amount of water and the amount of ice will decrease
- (5) Both the amount of water and the amount of ice will increase

41. Find the odd item in the thermodynamic process :
- (1) Pressure
 - (2) Temperature
 - (3) Density
 - ☒ (4) Volume
 - (5) Density and volume
42. For a non-flow constant pressure process, the heat exchange is equal to :
- (1) Zero
 - (2) The work done
 - (3) The change in internal energy
 - ☒ (4) The change in enthalpy
 - (5) The change in thermal energy
43. Identify the process of change of a close system in which the work transfer (expansion) is maximum :
- ☒ (1) Isothermal
 - (2) Isochoric
 - (3) Isentropic
 - (4) Polytropic
 - (5) Isobaric
44. The area under the temperature-entropy curve (T-s curve) of any thermodynamic process represents :
- (1) Heat absorbed
 - (2) Heat rejected
 - ☒ (3) Either heat absorbed or rejected
 - (4) Heat and work added
 - (5) Only heat added
45. An ideal gas of mass, m and temperature, T_1 undergoes a reversible isothermal process from an initial pressure p_1 to final pressure p_2 . The heat loss during the process is Q . The entropy change is :
- (1) $mR \ln \frac{p_2}{p_1}$
 - ☒ (2) $mR \ln \frac{p_1}{p_2}$
 - (3) $mR \ln \frac{p_2}{p_1} - \frac{Q}{T_1}$
 - (4) Zero
 - (5) $mR \ln \frac{p_1}{p_2} - \frac{Q}{T_1}$

46. The efficiency of Carnot cycle depends upon :

- ☒ (1) Temperature ratio
- (2) Pressure ratio
- (3) Volume ratio
- (4) Work ratio
- (5) Energy ratio

47. Choose the correct answer :

- (1) Specific volume of water decreases on freezing.
- (2) Boiling point of water decreases with increasing pressure.
- (3) Specific volume of CO_2 increases on freezing.
- ☒ (4) Freezing temperature of water decreases with increasing pressure.
- (5) Density of ice is minimum at 4 degrees celsius

48. Choose the correct answer :

- ☒ (1) The slope of vapourisation curve is always negative.
- (2) The slope of vapourization curve is always positive.
- (3) The slope of sublimation curve is negative for all pure substances.
- (4) The slope of fusion curve is positive for all pure substances.
- ☒ (5) The slope of fusion curve is negative for all pure substances.

49. Which of the following is an economic factor affecting the growth of entrepreneurship ?

- ☒ (1) Infrastructure
- (2) Religion
- (3) Family background
- (4) Value system
- ☒ (5) Social issues

50. An economic system may be :

- (1) Communist
- (2) Capitalist
- (3) Socialist
- (4) ~~Marxism~~
- (5) All of the above

51. Which of the following shows the process of creating something new ?

- (1) Business Model
- (2) Modeling
- (3) ~~Innovation~~
- (4) Creative flexibility
- (5) Creative model

52. Having less than 50% of equity shares in an international venture is called :

- (1) ~~Minority interest~~
- (2) Joint venture
- (3) Majority interest
- (4) Exporting
- (5) Bonding

53. The subject matter of Personnel Management is :

- (1) Profit
- (2) Capital Investment
- (3) ~~Human being~~
- (4) Wages and incentives
- (5) Asset and liability

54. Personnel policies are the _____ of the organisation.

- (1) ~~Constitution~~
- (2) Epic
- (3) Foundation
- (4) None of these
- (5) Principal values

55. The following is not the Principle of personal policies :

- (1) Principle of common interest
- (2) Principle of development
- (3) Principle of recognition of trade unions
- (4) ~~Principle of team spirit~~
- (5) Principle of flexibility

56. _____ means limitation of organised action.
- (1) Organising
 - (2) Controlling
 - ☒ (3) Directing
 - (4) Planning
 - (5) Implementing
57. Industry shift face is concerned with :
- (1) Motivate people
 - (2) Visualise picture
 - (3) Clear vision
 - ☒ (4) Comfortable with ambiguity
 - (5) Innovation and development
58. Which of the following is usually not an objective of Industrial relation ?
- (1) Connectedness
 - (2) Collective wisdom
 - (3) Conflict prevention
 - (4) Both (a) and (b)
 - ☒ (5) None of the above
59. That the authority rests solely with the management with no right to anyone to challenge, it is the basis of :
- (1) Pluralists approach
 - (2) System approach
 - ☒ (3) Unitary approach
 - (4) Social action approach
 - (5) Marxist approach
60. Identify the major actor of industrial relation from the following :
- (1) Employers
 - (2) Unions
 - (3) Government
 - (4) Organisation
 - ☒ (5) All of the above
61. Organisations are made up of people and the success of management lies in its dealings with these people, it is the fundamental of :
- (1) Gandhian approach
 - (2) Marxist approach
 - ☒ (3) Human relations approach
 - (4) Giri approach
 - (5) Communist approach

62. Which of the following is not typically a component of partnership agreement ?

(1) Commitment to information and consultation

☒ (2) Commitment to high levels of pay

(3) Sharing success with employed

(4) Commitment to employment security

(5) Commitment to sharing of profit

63. Entrepreneurship refers to a :

(1) Creation

☒ (2) Process

(3) Both (1) and (2)

(4) Creation and venture

(5) None of the above

64. For a damped harmonic oscillation motion, choose the correct answer :

(1) Amplitude as well as time period changes

(2) Only time period changes

☒ (3) Only amplitude changes

(4) Only wavelength changes

(5) All remain unchanged

65. What is the magnitude of dissipative force on an oscillator, when oscillator is at rest ?

☒ (1) Zero

(2) Infinity

(3) Any value

(4) Dependent on specific parameters

(5) Unpredictable

66. Which of the following properties does not change when wave passes from one medium to another ?

(1) Speed

(2) Wavelength

☒ (3) Frequency

(4) Amplitude

(5) Mass

67. In Newton's ring, the spacing between consecutive rings _____ with increase of order of rings.

- (1) Increases
- (2) Remains constant
- ☒ (3) Decreases
- (4) None of the above
- (5) May increase or decrease

68. Newton's ring are circular because the air space between plane glass plate and plano convex lens in _____.

- (1) Circularly non-symmetric
- ☒ (2) Circularly symmetric
- (3) Parabolic ally symmetric
- (4) Parabolic ally non-symmetric
- (5) Circularly antisymmetric

69. If light of wavelength is incident normal to the grating plane, the first order diffraction is observed at an angle θ , and N is the number of lines

per unit length of grating, then λ is _____.

- ☒ (1) $\sin\theta/N$
- (2) $N\sin\theta$
- (3) $N/\sin\theta$
- (4) $1/N\sin\theta$
- (5) $\frac{N}{\cos\theta}$

70. The refractive index of Canada balsam is :

- (1) Less than O-ray of calcite
- (2) Larger than E-ray of calcite
- (3) In between O-ray and E-ray of calcite
- ☒ (4) All of the above
- (5) None of the above

71. Which of the following is zero ?

- (1) Grad div
- (2) Div curl
- ☒ (3) Curl grad
- (4) Div grad
- (5) Curl div

72. If $\nabla \cdot \mathbf{D} = \epsilon \nabla \cdot \mathbf{E}$ and $\nabla \cdot \mathbf{J} = \sigma \nabla \cdot \mathbf{E}$ in a given material, the material is said to be :

- (1) Linear
- (2) Isotropic
- (3) Homogeneous
- ☒ (4) Linear and homogeneous
- (5) Linear and isotropic

73. Which of the following does not mean pointing vector ?

- (1) The amount of energy flowing per second per unit area
- ☒ (2) The amount of power flowing per second per unit area
- (3) The amount of power flowing per unit area
- (4) The amount of power flowing per second
- (5) The amount of energy flowing per second

74. Which of the following is/are not the property / properties of electromagnetic waves ?

- (1) Phase of plane electromagnetic wave is constant throughout the propagation
- (2) Wave front of a plane electromagnetic wave is plane
- (3) Propagation vector is perpendicular to the wave front
- ☒ (4) The plane electromagnetic wave has not single frequency
- (5) Both (1) and (2)

75. Name a phenomenon where energy is converted into matter :

- (1) Compton effect
- (2) Thompson effect
- (3) Radioactive decay
- ☒ (4) Pair production
- (5) Raman effect

76. What is the unit of two dimensional wave function ?

- (1) Meter^{-3/2}
- ☒ (2) Meter⁻¹
- (3) Meter^{-1/2}
- (4) Meter⁻²
- (5) None of the above

77. As per quantum physics, the energy of a free particle is :

- (1) Quantized
- ☒ (2) Continuous
- (3) Always zero
- (4) Always infinity
- (5) Always constant

78. What is the unit of transmission coefficient ?

- ☒ (1) Unit less
- (2) Weber/meter
- (3) Tesla
- (4) Candela
- (5) Mole

79. If either the inductance or the rate of change of current is doubled, the induced EMF :

- (1) Remains constant
- (2) Becomes zero
- ☒ (3) Doubles
- (4) Becomes half
- (5) Becomes four times

80. If 1 A current flows in a circuit, the number of electrons flowing through this circuit is :

- ☒ (1) 0.625×10^{19}
- (2) 1.6×10^{19}
- (3) 0.625×10^{-19}
- (4) 1.6×10^{-19}
- (5) 1.625×10^{18}

81. The resistivity of the conductor depends on :

- (1) Area of the conductor
- (2) Length of the conductor
- ☒ (3) Type of the material
- (4) Volume of the conductor
- (5) Dimensions of the conductor

87. Resistance of a VARISTOR varies with :
- (1) Voltage
 - (2) Current
 - (3) Temperature
 - (4) Both (1) and (3)
 - (5) Both (2) and (3)
88. Eddy current loss will depend on :
- (1) Frequency
 - (2) Flux density
 - (3) Thickness
 - (4) All of the above
 - (5) None of the above
89. A DC generator without commutator is a :
- (1) AC generator
 - (2) DC motor
 - (3) DC generator
 - (4) Induction motor
 - (5) AC motor
90. In a DC machine, if 72 number of coils are used, what is the number of commutator segments required ?
- (1) 36
 - (2) 34
 - (3) 72
 - (4) 74
 - (5) 84
91. Lap winding is preferred for which type of machines ?
- (1) High current and low voltage
 - (2) High current and high voltage
 - (3) Low current and low voltage
 - (4) Low current and high voltage
 - (5) Both (3) and (4)
92. A shunt generator running at 1000 rpm has generated emf of 100 V. If the speed increases to 1200 rpm, the generated emf will be nearly :
- (1) 140 V
 - (2) 120 V
 - (3) 175 V
 - (4) 220 V
 - (5) 250 V

93. Five cells are connected in series in a row and then four such rows are connected in parallel to feed the current to a resistive load of 1.25Ω . Each cell has an emf of 1.5 V with internal resistance of 0.2Ω . The current through the load will be :

- (1) 3.33 A
- (2) 23.33 A
- (3) ~~5 A~~
- (4) 1 A
- (5) 2 A

94. Application of DC voltage to a diode, transistor, or other device to produce a desired mode of operation is called :

- (1) ~~Biasing~~
- (2) Reduction
- (3) Bounding
- (4) Modulation
- (5) Saturation

95. Group of eight bits in a binary data is called :

- (1) Nibble
- (2) Bitwidth
- (3) ~~Byte~~
- (4) Word
- (5) Kilobyte

96. Amount of energy required to produce full conduction across p-n junction in forward bias is called :

- (1) ~~Barrier potential~~
- (2) Barrier difference
- (3) Barrier intensity
- (4) Barrier density
- (5) Barrier bias

97. Value of gate to source voltage in FET that makes drain current approximately zero is :

- (1) ~~Cutoff voltage~~
- (2) Saturation voltage
- (3) Biasing voltage
- (4) Source voltage
- (5) Reverse voltage

98. 1011000011010000 to hexadecimal number is :

(1) 1010

☒ (2) B0E0

(3) F0F0

(4) 10F0

(5) B0F0

99. If collector current is 2 mA and base current is 0.5 mA, than emitter current will be :

(1) 10 mA

☒ (2) 2.5 mA

(3) 1.5 mA

(4) 4 mA

(5) 2 mA

100. In saturation region, base resistor of general BJT model will be $10\text{k}\Omega$ with base supply voltage of 3V than collector current will be :

(1) 0.32 mA

☒ (2) 0.23 mA

(3) 2.23 mA

(4) 3.32 mA

(5) 3.23 mA

